Cluster-Level Weighted SAR Radiometric Cross-Calibration Using Urban Building Areas

Yingkui Zhang
College of Electronics and Information Engineering
Shenzhen University
Shenzhen, China
mgwdz_1122@163.com

Lei Huang*
State Key Laboratory of Radio Frequency Heterogeneous Integration
Shenzhen University
Shenzhen, China
lhuang@szu.edu.cn

Qiuchen Liu*
College of Electronics and Information Engineering
Shenzhen University
Shenzhen, China
liuqiuchen1014@163.com

Guitong Chen
College of Electronics and Information Engineering
Shenzhen University
Shenzhen, China
aries_chengt@126.com

***Abstract*—Accurate radiometric calibration of synthetic aperture radar (SAR) is a prerequisite for quantitative remote sensing applications. Conventional cross-calibration approaches based on natural pseudo-invariant features (PIFs) are susceptible to environmental variations such as wind speed, humidity, and seasonal effects, which will introduce considerable uncertainty. This work proposes a new SAR radiometric cross-calibration method based on feature weights allocation in building areas. By employing a multi-factor joint weighting model, the proposed method extracts high-stability radiometric features from building clusters, and then estimates the cross-calibration constant to correct the target images. Experimental results demonstrate that the proposed method achieves a calibration accuracy of 0.15 dB. It eliminates the need for reflectors/active calibrators and dedicated calibration sites, enabling low-cost and high-frequency calibration updates.**

***Index Terms*—radiometric cross-calibration, synthetic aperture radar (SAR), building areas, feature weights allocation.**

I. Introduction

Synthetic Aperture Radar (SAR) is an active microwave imaging system that provides continuous observations independent of illumination and most weather conditions. It enables the stable acquisition of surface backscatter information even under complex meteorological conditions, such as cloud cover and precipitation. Consequently, SAR has become a vital data source for Earth observation, extensively supporting applications including surface process analysis, sea-ice and land-ice monitoring, and disaster assessment [1]–[3]. To facilitate quantitative remote sensing applications and data fusion across diverse regions, platforms, and imaging modes, it is imperative to ensure the radiometric consistency and comparability of different SAR products. Accordingly, rigorous radiometric calibration and long-term radiometric quality maintenance are essential requirements [4]–[6].

This work was supported by the National Key R&D Program of China under Grant 2023YFB3905500.

From a methodological perspective, traditional SAR radiometric calibration primarily encompasses calibration based on artificial benchmarks (e.g., corner reflectors and active transponders) and external field validation utilizing typical stable natural scenes [7]–[9]. While artificial calibrators provide reference scattering characteristics with high confidence, they are constrained by high deployment and maintenance costs as well as limited spatial coverage, making it difficult to satisfy the requirements of high-revisit frequency and multi-mode missions [10], [11]. Simultaneously, the scattering properties of distributed natural scenes are significantly susceptible to climatic conditions and seasonal variations, posing challenges in guaranteeing their long-term radiometric stability.

Radiometric cross-calibration significantly mitigates the reliance on calibration artifacts while demonstrating superior timeliness and scalability by leveraging a well-calibrated reference sensor to perform radiometric correction on a target sensor [12], [13]. Traditional cross-calibration strategies primarily rely on natural Pseudo-Invariant Features (PIFs) or Pseudo-Invariant Calibration Sites (PICS), establishing quantitative relationships between reference and target images by selecting land-cover regions with stable radiometric time series [14], [15]. However, natural PIFs and PICS are frequently affected by environmental perturbations, such as wind speed, humidity, and seasonal variations. Furthermore, their stability exhibits significant spatio-temporal dependence, leading to increased estimation uncertainty. In this context, urban building areas have emerged as promising candidates for alternative natural calibration units due to their intense backscattering and relatively stable structural morphology.

In [16], the temporal stable building slices were selected via neural networks, and their median centroid was utilized as calibration benchmark to achieve periodic monitoring of absolute calibration constants. In [17], building pixels were extracted using sliding windows and land cover classification, and stable pixels were further screened by combining scattering effect thresholds. Subsequently, a calibration reference was constructed using the global mean to complete cross-calibration. However, the aforementioned methods oversimplify the entire building areas by treating it as a homogeneous target, neglecting the internal variations in scattering intensity and stability across different structures.

In this work, we propose a SAR radiometric cross-calibration method based on feature weights allocation in building areas. Our contributions can be summarised as the following:

- We consider the building areas as PIFs and explicitly model the spatial heterogeneity induced by fragmented building clusters, thereby effectively suppressing outlier scattering and scene non-uniformity.
- We introduce a multi-factor joint weighting method to reduce sensitivity to hyperparameter choices and improve the estimation accuracy of stable radiometric features.
- The proposed method eliminates the need for corner reflectors and active calibrators, enabling a streamlined workflow with low cost, wide-area coverage, and high operational scalability.

II. Method

A. Extraction and Clustering of Building Areas

Cross-calibration achieves radiometric consistency by establishing a quantitative relationship between calibrated and uncalibrated images.To ensure the comparability of backscattering intensities across different SAR images, the images must be unified within a consistent spatial and radiometric framework. The selected calibrated and uncalibrated SAR images are required to share an identical geographic projection and spatial resolution grid. Furthermore, to minimize systematic discrepancies, the acquisition parameters—specially local incidence angles, imaging modes, frequency bands, and product levels—must remain consistent between the two datasets.

In this paper, the calibrated SAR image is designated as the reference image, denoted as I_r, while the uncalibrated SAR image is designated as the target image, denoted as I_t. To mitigate non-target discrepancies induced by imaging geometry and system noise, we apply standardized preprocessing to both SAR images I_r and I_t. This preprocessing jointly corrects orbit and terrain-related geometric distortions and reprojects the imagery into a map coordinate system, and mitigates radiometric perturbations such as additive thermal noise from the imaging chain and coherent speckle noise.

In the following, the backscattering coefficient is adopted as the radiometric parameter. Let the pixel sets over building areas in the reference and target images be denoted by $\mathbf{\Omega}_r$ and $\mathbf{\Omega}_t$ with corresponding pixel values $x_r(p)$ and $x_t(p)$, where p denotes the pixel point. The objective of cross-calibration is to estimate a calibration constant $\mathcal{L}$ such that the corrected target image I_t' is radiometrically consistent with the reference image I_r. The relationship can be expressed as [17]

$$x_r(p)_{\text{dB}} \approx x_t'(p)_{\text{dB}} = x_t(p)_{\text{dB}} + \mathcal{L}_{\text{dB}}. \tag{1}$$

To enable a more accurate extraction of building areas pixels, we incorporate an external land cover product to construct an initial building areas mask, and project it onto a co-registered raster grid to obtain the building areas pixels set $\mathbf{\Omega}$. Although it provides a coarse delineation, the resulting pixels set may still contain exceptionally strong scatterers and mixed pixels that can bias mean-based statistics. To mitigate their influence, we perform percentile-based clipping on the pixels values. Let $[q_\alpha, q_\beta]$ be the lower and upper percentile thresholds of building-pixel values, then the retained set is

$$\mathbf{\Omega}^c = \{p \in \mathbf{\Omega} \mid q_\alpha \le x(p) \le q_\beta\}. \tag{2}$$

By utilizing (2), the clipped building areas pixels sets $\mathbf{\Omega}_r^c$ and $\mathbf{\Omega}_t^c$ for the reference and target images are obtained, respectively. The purpose is to improve the robustness of subsequent clustering and weighted estimation by reducing sensitivity to extreme outliers.

Urban building areas exhibit multi-scale spatial patterns, such as dense downtown cores, moderately dense residential zones, and sparse industrial/suburban regions. To capture spatial heterogeneity, Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is employed to cluster building pixels based on spatial density connectivity [18]. DBSCAN can identify core points by defining a neighborhood radius Eps and a minimum number of points $MinPts$.

Let $\mathbf{u}(p) = [\text{row}(p), \text{col}(p)]^T$ denote the coordinate of pixel p in the SAR image. Given Eps and $MinPts$, the distance between two pixels p_i and p_j is defined as the Euclidean distance

$$d(p_i, p_j) = \|\mathbf{u}(p_i) - \mathbf{u}(p_j)\|_2 . \tag{3}$$

Accordingly, the Eps-neighborhood of p is

$$\mathcal{N}(p) = \{p, \bar{p} \in \mathbf{\Omega}^c \mid d(p, \bar{p}) \le Eps\}, \tag{4}$$

and a pixel is regarded as a core point if $|\mathcal{N}(p)| \ge MinPts$. By expanding clusters through density-reachable relationships and designating the remaining points as noise, DBSCAN can discover arbitrarily shaped clusters without prior knowledge of the number of clusters.

Therefore, we can partition $\mathbf{\Omega}^c$ into L clusters $\{\mathcal{C}_l\}_{l=1}^L$ and a small set of noise points. Each cluster $\mathcal{C}_l$ corresponds to a building ensemble that is treated as a candidate local PIF unit within the cross-calibration imaging geometry.

B. Clusters Stable Features and Multi-Factor Joint Weighting

Given the l-th building cluster $\mathcal{C}_l$ of size N_l, its cluster mean in the reference and target images is determined as

$$\mu_{r,l} = \frac{1}{N_{r,l}} \sum_{p \in \mathcal{C}_{r,l}} x_r(p), \quad \mu_{t,l} = \frac{1}{N_{t,l}} \sum_{p \in \mathcal{C}_{t,l}} x_t(p). \tag{5}$$

Unless stated otherwise, subscripts r and t represent the reference and target images, respectively. The within-cluster dispersion (standard deviation) v_l is used to characterize radiometric consistency. To mitigate the bias associated with small-sample clusters, v_l is defined as the sample standard deviation

$$v_l = \sqrt{\frac{1}{N_l - 1} \sum_{p \in \mathcal{C}_l} (x(p) - \mu_l)^2}. \tag{6}$$

Smaller v_l indicates higher within-cluster stability. However, the within-cluster dispersion alone is insufficient to serve as a robust proxy for radiometric confidence. While DBSCAN performs a binary identification of clusters, a cluster density metric is introduced to achieve a more robust evaluation of cluster credibility and quality. The density ρ_l is estimated using a K-Nearest Neighbor (KNN) scheme [19]. For each pixel in the cluster, let $d_k(p)$ denote the mean distance to its k-th nearest neighbor. The ρ_l can be given by

$$\rho_l = \frac{1}{(N_l - 1)(d_k(p) + \epsilon)}, \tag{7}$$

where ϵ is a small constant for numerical stability.

Let $\sigma_l = 1/v_l$, the parameters μ_l, σ_l and ρ_l are normalized to [0,1] via the min-max scaling method, defined as $\bar{\mu}_l$, $\bar{\sigma}_l$ and $\bar{\rho}_l$. A multi-factor joint weight is then constructed as

$$\omega_l = \sqrt{\bar{\mu}_l^2 + \bar{\sigma}_l^2 + \bar{\rho}_l^2}, \tag{8}$$

The normalized weight is

$$\bar{\omega}_l = \frac{\omega_l}{\sum_{l=1}^{L} \omega_l}. \tag{9}$$

The multi-factor joint weight ensures that larger size, higher density, and more consistent clusters contribute more to the final calibration constant, aligning with the principle that higher-confidence PIF units should dominate the estimate.

C. Radiometric Cross-Calibration

At the cluster level, the dB-domain difference observation is defined as

$$\delta_l = \mu_{r,l} - \mu_{t,l}. \tag{10}$$

Assuming the cross-calibration constant is spatially invariant within the region of interest, the global constant is estimated via weighted aggregation

$$\mathcal{L} = \sum_{l=1}^{L} \bar{\omega}_l \delta_l. \tag{11}$$

The target image is then corrected by (1). Compared with whole building areas averaging or single natural PIF estimation, the proposed cluster-level weighted strategy suppresses outlier and unstable clusters, leading to reduced variance and improved robustness of $\mathcal{L}$.

III. Experimental Results and Analysis

A. Data and Experimental Setup

Experiments were conducted using Sentinel-1 SAR data from the European Space Agency's Sentinel-1 satellite. To better evaluate the impact of urban building distribution on the proposed algorithm's performance, Los Angeles, USA, was selected as the experimental area. The specific terrain distribution and major building coverage are shown in Fig. 1. To provide a preliminary validation of the structural invariant properties of urban building areas, a total of 6 SAR images acquired between April and July 2019 were selected as target and reference images. Specifically, three images were designated as target SAR images to calculate their individual calibration constants, while the remaining three images served as reference SAR images, with the mean of their calibration constants used as the theoretical value. All data were processed using the Sentinel-1 processing workflow in SNAP to ensure radiometric and geometric comparability. For the reference images, the pipeline included orbit correction (optional), thermal noise removal, radiometric calibration, and terrain correction. For the target images, orbit correction, thermal noise removal, and terrain correction were applied. After processing, both images were mapped to a common grid and coordinate frame, providing comparable backscattering coefficient measurements for subsequent feature extraction and cross-calibration.

B. Experiment of SAR Radiation Cross-Calibration

Building areas are extracted using an external land-cover product provided by ESA, which was generated from Sentinel-1/Sentinel-2 data and aligned to a 10m grid [20]. The building pixel values are filtered using 2nd-98th percentile thresholds to ensure data quality. The SAR image of the experimental area, following this initial filtering process, is presented in Fig. 2. Subsequently, DBSCAN is employed to partition the trimmed building pixels into density-connected building clusters. The global DBSCAN parameters are set to $Eps = 30$ and $MinPts = 50$. Notably, an adaptive selection strategy for varied urban morphologies can be implemented by scaling these parameters relative to the spatial resolution and the clustering result of building areas.

Fig. 3 illustrates the DBSCAN clustering result for a SAR image. It can be observed that after filtering out noise and micro-clusters, four valid clusters are identified. By treating these as four stable targets, varying weights can be assigned

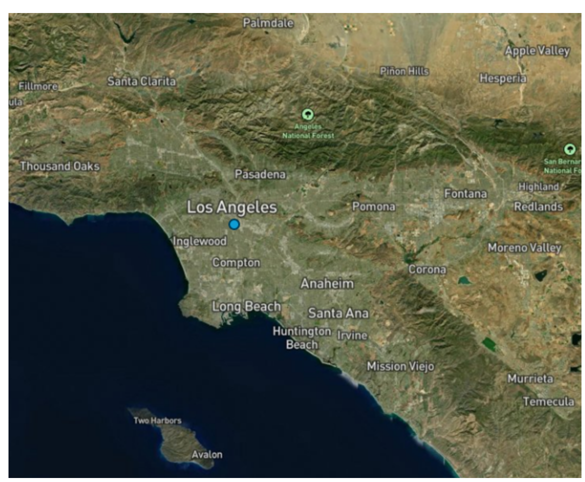

Fig. 1. Specific terrain distribution and major building coverage of Los Angeles.

based on their size, dispersion, and density, thereby forming an integrated feature for radiometric energy extraction. Then, after computing the normalized weights $\bar{\omega}_{r,l}$ and $\bar{\omega}_{t,l}$ for the reference and target SAR datasets by (9), the calibration constant $\mathcal{L}$ is further determined. The final calibration accuracy is determined by evaluating the deviation of $\mathcal{L}$ from the theoretical calibration constant $\mathcal{L}_0$. According to the Sentinel-1 lookup table (LUT), the theoretical value is $\mathcal{L}_0 = 27.758$ dB. Furthermore, the Root Mean Square Error (RMSE) is employed as metric to quantify the calibration accuracy, as defined by the following formula

$$\mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(\mathcal{L}_t - \mathcal{L}_0)^2}, \tag{12}$$

where T denotes the number of target SAR images.

To evaluate the effectiveness of the weighting allocation strategy, the proposed method in this paper is compared against the non-clustered and statistics approach that uses the equal-weighted value (mean) of pixels as the benchmark. The compared method was employed to analyze the calibration performance in [17].

Table I presents the SAR radiometric cross-calibration results based on feature weights allocation and the equal-weighted allocation in building areas. The equal-weighted baseline represents the standard operational logic of traditional PIF-based calibration. By transitioning the modeling of building areas from a traditional homogeneous entity to a heterogeneous cluster-level weighted framework, varying weights are assigned to distinct clusters based on their spatial distribution characteristics and radiometric quality parameters. Meanwhile, the calculated calibration constants closely approximate the theoretical values $\mathcal{L}_0$ in our method, and the RMSE is reduced from 0.36 dB to 0.15 dB. It can be observed that by explicitly characterizing spatial heterogeneity via clustering and employing a multi-factor joint weighting strategy, the proposed method achieves superior calibration accuracy in dispersed building regions.

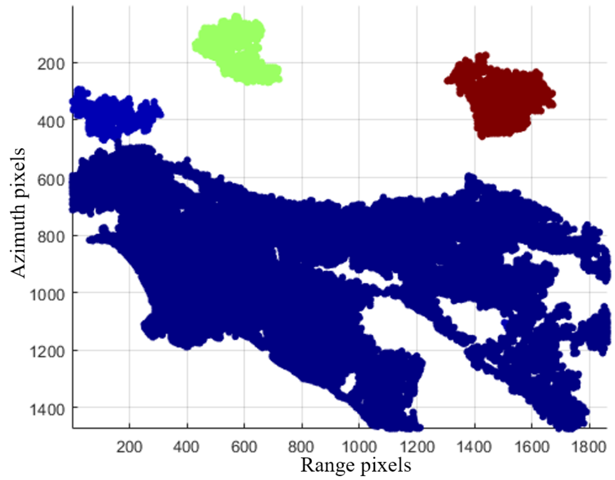

Fig. 3. Clustering result of the DBSCAN algorithm.

TABLE I
RADIOMETRIC CROSS-CALIBRATION RESULT (dB)

Method	**Image label**	**Weight**	**Calibration constants**	**Calibration accuracy**
Proposed	1	[0.458, 0.196, 0.214, 0.131]	27.955	RMSE = 0.152
	2	[0.550, 0.149, 0.181, 0.121]	27.865	
	3	[0.602, 0.141, 0.146, 0.112]	27.897	
Equal-weighted	1	[0.25, 0.25, 0.25, 0.25]	28.339	RMSE = 0.365
	2		27.942	
	3		27.925	

Fig. 2. Building areas extraction result for the SAR image.

IV. CONCLUSION

This paper presents a feature weights allocation method for SAR radiometric cross-calibration. By integrating density-based clustering and a joint weighting scheme incorporating scale, density, and dispersion, the structural heterogeneity of building areas is explicitly accounted for in the estimation of calibration constants. Consequently, this framework statistically mitigates estimation bias and variance. Experimental results based on Sentinel-1 datasets demonstrate high consistency between the derived calibration constants and theoretical benchmark, with the calibration accuracy maintained at approximately 0.15 dB. This represents a performance enhancement over the traditional unclustered building pixels-mean baseline.

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